James Newton Leathers IV
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Education

Purdue University - West Lafayette, IN | Major: **Electrical Engineering Technology** | Graduation: May 2026
Minor: **Organizational Leadership**

Work Experience

Electronics Hardware Engineer & PCB Design Intern | Hughes Network Systems, LLC & EchoStar | Summer 2025

- Reverse engineering legacy embedded system hardware for modernization integration with Cadence Allegro, Blueprint, & Viewmate Software; presented completed technical documentation to company SVPs & VPs.

RF Applications Undergrad Lab Teaching Assistant | Wireless Communications | 2025 - Present

- Instructing students how to build & conduct analysis of a superheterodyne receiver & SMT components.

Software Undergrad Lab Teaching Assistant | Intro to Digital Systems | 2024 - 2025

- Instructed classes utilizing Microchip Studio C programming for weekly ATMEGA 2560-based projects.

Mentor - High School Programming Project | Spring 2024

- Mentored 9th grade high school student in C programming for ATMEGA 328p based science project.

Handyman & Landscaping Assistant | J Handyman, LLC | 2018 - 2021

- Assisted in minor electrical & plumbing repairs; operated machinery on commercial & residential properties.

Platforms

Cadence Allegro | Hardware Engineering/PCB Design Internship

- Designed embedded system hardware at Hughes Network Systems, LLC under NDA.

Docker | Personal Projects

- Deployed AI Surveillance & Smart Automation containers for a real-time local network server.

Node-RED | Computer Based Data Acquisition Applications

- Developed data acquisition systems with Raspberry Pi 4 Linux OS, JavaScript, & Python.

Microchip Studios & STM32CubeIDE | Advanced Embedded Digital Systems

- Designed ARM Cortex-M4 based systems with STM32L476RG & used ATMEGA 2560 & 328P Microcontrollers.

Software

- MATLAB Simulink (Digital Signal Processing – Analyzing digital signal fundamentals)
- NI Multisim (DC & AC Power Electronics - simulation, prototypes, & calculations)
- SOLIDWORKS & Fusion (Adv. Embedded Digital Systems – Designed Components for 3D Print implementation)
- KiCad & EAGLE (Electronic Prototype Development & Adv. Embedded Digital Systems – Custom PCBs)
- Quartus II (Concurrent Digital Systems - FBGA board AHDL programming)

Projects

Industry Sponsored Senior Capstone | Cummins Inc. & Purdue Polytech Institute | 2025 - Present

- Team project manager working with AI & machine learning for a deliverable product for a client under NDA.

AI Surveillance & Smart Automation Server

- Developed Linux-based local network server utilizing Docker to host an AI-powered NVR for intelligent video surveillance with YAML; integrated Home Assistant, MQTT Broker, & Zigbee comm. protocol for IoT stack automation.

Spot Micro Quadruped Robot

- Designing & currently building an open-sourced AI-piloted Spot Micro Robot Dog, inspired by Boston Dynamics' Spot; implementing computer vision, RF control, & embedded control systems.

Signal Hacking

- Exploring NFC & 125kHz RFID signal emulation with Flipper Zero.

Stereo Audio Amplifier +10W (University)

- Worked in a team environment to develop an embedded digital system for a jukebox project.

Superheterodyne FM Radio Receiver (University)

- Developed & tested subcircuits in wireless communications to build an FM radio with Class A amplifier.

Programmable Crane (University)

- Worked in a team environment to develop EEPROM programmable crane.

Extracurriculars

Phi Kappa Tau Fraternity Member | 2024 - Present

- Executive Board Membership Orientation Officer (Implemented member education program) | 2024

Tae Kwon Do – 2nd Degree Black Belt Kukkiwon Dan Certificate (United States Martial Arts Academy) | 2009 - 2019

Certifications

CPR/AED For the Professional Rescuer Certificate (Purdue University & American Red Cross) | 2024 - Present
Jr. Open Water Diver Certificate (Professional Association of Diving Instructors | PADI) | 2015 - Present
Library of Congress Researcher | 2024 - Present

Foreign Language

German - 3 years of study